

Mucus-Inspired Underwater-Enhanced Amphiphilic Liquid-Like Fluorocarbon Composite Coating: Antifouling and Anti-icing Cycle Durability

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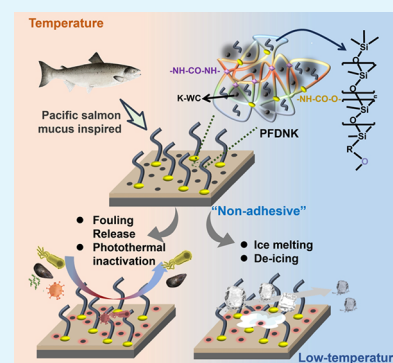
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ABSTRACT: The adhesion of marine fouling organisms and wave-induced ice accretion poses significant challenges to marine infrastructure, leading to substantial economic losses. Inspired by the Pacific salmon mucus, which exhibits both antibacterial and antifreeze properties, an amphiphilic liquid-like fluorocarbon composite coating (PFDNK) with antifouling/anti-icing cyclability and durability is fabricated by integrating thiol-modified tungsten carbide (K-WC) as cross-linking sites into a polyurethane/polyurea-modified fluorocarbon polymer. Remarkably, PFDNK (underwater coefficient of friction ~ 0.1) significantly reduces biofouling adhesion ($98.6 \pm 6.1\%$ inhibition ratio against *Pseudomonas xiamenensis*) and ice adhesion strength ($\sim 10.0 \pm 0.9$ kPa after 20 icing/deicing cycles). The presence of multiple hydrogen bonds within PFDNK effectively enhances its mechanical properties, achieving a tensile stress of 12.9 MPa and an elongation at break of 608%. Furthermore, under solar irradiation, K-WC endows the coating with photothermal properties, enabling exceptional contact-killing efficacy and ice-melting capability. PFDNK demonstrates outstanding frost- and snow-resistant performance under real-world low-temperature conditions, maintains excellent antifouling durability for approximately 240 days in marine environments, presents a promising solution for marine equipment spanning temperate and low-temperature zones, and expands applications in the marine industry.

KEYWORDS: antifouling, anti-icing, liquid-like, biomimetic material, fluorocarbon



1. INTRODUCTION

Underwater equipment, such as ships, frequently suffers from the adhesion of various types of fouling, leading to a series of issues,^{1,2} including threats to crew safety, the shortened service life of equipment, and biological invasions.³ Simultaneously, for vessels navigating across different maritime regions, ice formation in low-temperature waters not only accelerates the failure of antifouling coatings but also causes ice accumulation on the hull surface, resulting in additional economic losses.^{4,5} In winter, parts of the Arctic and the Southern Ocean freeze, and certain areas of China's Bohai Sea and Yellow Sea also freeze. For example, in the Liaodong Bay area, large shellfish fouling organisms such as mussels thrive in summer, but in winter, drift ice spanning dozens of nautical miles may form.^{6,7} Therefore, ships navigating in this sea area would better protect their hulls and reduce human and financial costs if coated with coatings that combine antifouling and anti-icing properties. The Pacific salmon is an anadromous fish capable of migrating between frigid freshwater rivers and temperate marine environments.^{8,9} Its surface mucus exhibits antibacterial, friction-reducing, and antifreeze properties, enabling adaptation to such alternating warm and cold conditions. Inspired by the Pacific salmon mucus, the development of a

multifunctional antifouling/anti-icing coating presents a promising solution for ships (or other underwater equipment) navigating between temperate and low-temperature regions to address severe biofouling challenges and mitigate the adverse effects of sea ice adhesion.

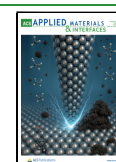
Owing to their low surface energy and absence of antifouling agent release, fluorocarbon compounds have emerged as a prominent class of applied antifouling coatings, attracting extensive research interest.¹⁰ Fluorocarbon compounds exhibit the capability to effectively fouling release under external shear forces.¹¹ However, the poor mechanical properties and weak adhesion significantly limit their applications in marine environments.¹² Recent studies have demonstrated that polyurethane/polyurea modification effectively enhances adhesion strength,¹³ while also enabling the grafting of antifouling agents to improve static antifouling performance.^{14,15} An

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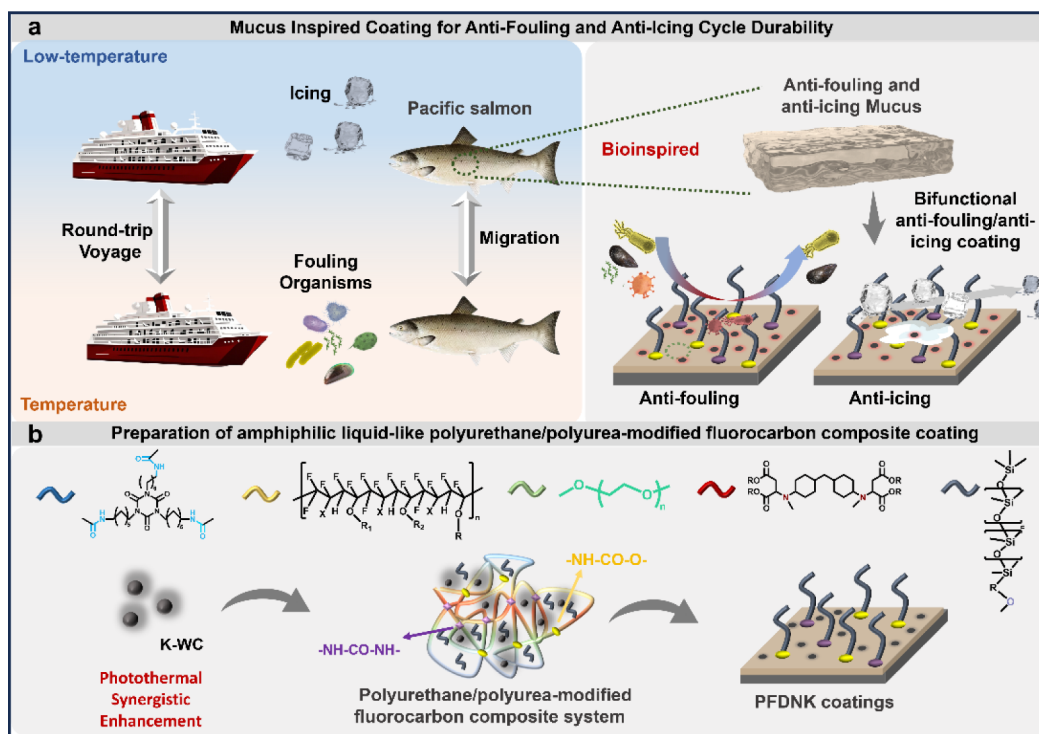
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Scheme 1. Schematic Illustration of the Pacific Salmon-Inspired Coating for Antifouling and Anti-icing: (a) Scheme Diagram of the Dual Antifouling and Anti-icing Mechanism; (b) Schematic Illustration of the Coating Preparation Process (Gray Lines Represent PDMS–OH, Blue Lines HDIT, Yellow Lines FEVE, Green Lines PEG, and Red Lines NH-1420)



additional advantage of this approach is that it enables the cross-linking of polymer segments with different characteristics (e.g., hydrophilic polymers, flexible polymer chains), thereby modifying the surface properties of fluorocarbon compounds to form amphiphilic coatings with both antifouling and anti-icing capabilities.^{16,17} Guo et al.¹⁸ introduced poly(*N*-vinylpyrrolidone) into polydimethylsiloxane (PDMS) to develop a versatile amphiphilic surface with bactericidal properties; similarly, Hu et al.¹⁹ fabricated a polyurethane-based polymer cross-linked with fluorocarbon and polyethylene glycol as amphiphilic segments, which exhibited antifouling, self-healing, and strong substrate adhesion capabilities. A notable example is the work by Mossayebi et al.,²⁰ who developed an amphiphilic poly(ionic liquid) surface based on PDMS and oligo(ethylene glycol) methyl ether methacrylate, achieving remarkable anti-icing effects. The incorporation of flexible chain segments into amphiphilic systems enables the formation of a smooth, liquid-like surface, which effectively inhibits biofouling adhesion.²¹ Concurrently, the hydrophobic components within the system reduce ice adhesion strength, while the hydrophilic segments delay ice nucleation.²² Therefore, the amphiphilic liquid-like surface of the polyurethane/polyurea-modified fluorocarbon polymer achieves simultaneous antifouling and anti-icing performance, enabling its application in fluctuating-temperature marine environments.

Additionally, introducing inorganic materials into amphiphilic liquid-like systems can enhance the cross-linking density of polymers²³ and synergistically meet the antifouling and anti-icing requirements of underwater equipment in complex marine environments.²⁴ Introducing photothermal inorganic materials into amphiphilic systems is generally an effective strategy. Through the photothermal properties of these materials can endow the coating surface with ice-melting

capabilities;²⁵ simultaneously, materials with high photothermal efficiency exhibit excellent antibacterial properties.^{26,27} Transition metal carbides possess ideal photothermal conversion efficiency and satisfactory thermal stability.^{28,29} Among these, tungsten carbide (WC) stands out as a promising candidate for solar absorbers due to its strong photothermal conversion capability,^{30,31} its potential for anti-icing and antifouling applications remains largely unexplored. Given WC's exceptional photothermal capabilities, we contend that it holds significant promise for advanced ice-phobic and antifouling surface development.

Inspired by the dual antimicrobial/anti-icing properties of Pacific salmon mucus, we design a bifunctional antifouling/anti-icing coating for temperate and low-temperature environments (Scheme 1a), mimicking the antibacterial, friction-reducing, and ice-resistant surface characteristics of Pacific salmon mucus. The coating matrix consists of a polyurethane/polyurea-modified fluorocarbon system formed through cross-linking fluoroethylene vinyl ether (FEVE) copolymer as the rigid segment, polyethylene glycol (PEG) as the hydrophilic segment, and hexamethylene diisocyanate trimer (HDIT) as the bridging agent. Meanwhile, monohydroxyl-terminated silicone oil (PDMS–OH) is introduced as the flexible segment, and polyaspartic acid ester (NH-1420) is used as the curing agent to balance flexibility and rigidity. PDMS–OH enriches at the coating surface to impart low-adhesion properties and underwater friction reduction. Photothermal thiol-modified tungsten carbide (K-WC) is introduced as an inorganic structural unit to provide additional cross-linking sites, not only enhancing cross-link density to minimize structural defects, but also endowing the coating with photothermal conversion capability—thus enabling dual-function photothermal sterilization and ice-melting capacities,

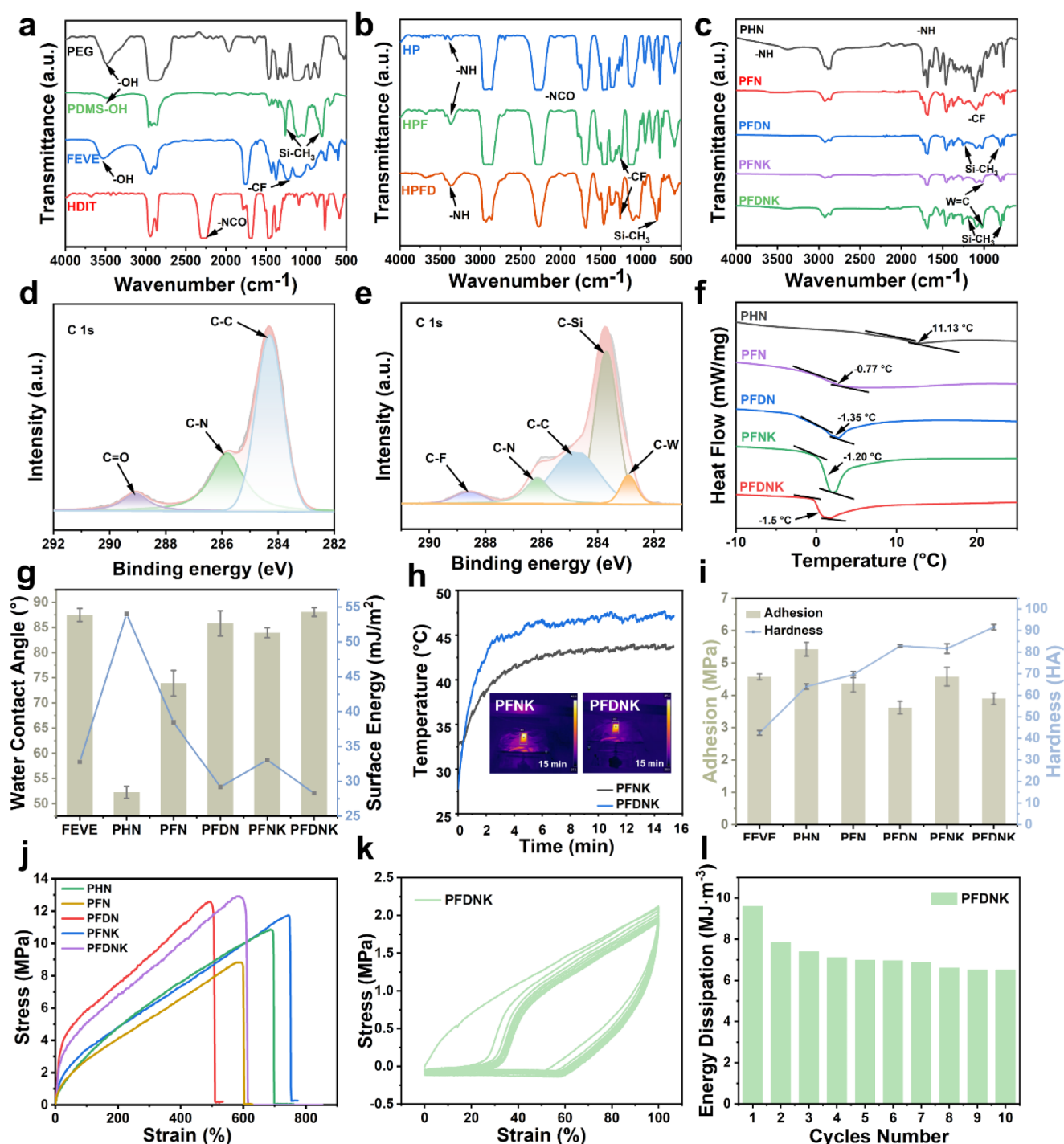


Figure 1. Chemical characteristics of the PHN, PFN, PFDN, PFNK, and PFDNK. FTIR spectra of (a) PEG, PDMS-OH, FEVE, HDIT, (b) HP, HPF, HPFD, (c) PHN, PFN, PFDN, and PFNK. High-resolution C 1s XPS spectra of (d) PHN, (e) PFDNK. (f) The DSC pattern. (g) Water contact angle and surface energy. (h) PFNK and PFDNK temperature change during 15 min, with an inset showing thermal infrared image of PFNK and PFDNK. (i) Hardness and adhesion. (j) Stress-strain curve. (k) Cyclic tensile test of PFDNK. (l) The energy dissipation corresponds to the cyclic tensile test of PFDNK.

ultimately forming the amphiphilic liquid-like polyurethane/polyurea-modified fluorocarbon composite coating (PFDNK) shown in Scheme 1b. (Note: In Scheme 1, gray lines represent PDMS-OH, blue lines HDIT, yellow lines FEVE, green lines PEG, and red lines NH-1420.) The nonadhesive properties and underwater friction-reducing capability (coefficient of friction ~ 0.1) of PFDNK significantly reduce the adhesion of fouling organisms and maintain low ice adhesion strength even

after 20 icing/deicing cycles. Under light irradiation (AM 1.5), the synergistic effect of the amphiphilic liquid-like surface and photothermal K-WC contributes to strong antibacterial performance and efficient ice-melting capability. Moreover, in real-world marine applications (temperature range: -15 to 15 °C), the coating demonstrates exceptional durability, maintaining high performance for approximately 240 days, highlighting its practicality and feasibility in shallow marine

environments. Even under harsh simulated environments, including high-salinity immersion and turbulent flow conditions, the PFDNK consistently demonstrates robust surface stability. This coating introduces a novel antifouling strategy and fabrication methodology, offering a promising solution for protecting the surfaces of ships and marine infrastructure operating in marine environments with varying temperature regimes.

2. RESULTS AND DISCUSSION

2.1. Synthesis Strategy and Characterization. The synthesis of the fluorocarbon composite coating involves the preparation of prepolymers, polyurethane/polyurea-modified fluorocarbon polymer, and the photothermal material K-WC, as illustrated in Figures S1–S3. A series of prepolymers (HP, HPF, and HPFD) is prepared by adjusting the content of PDMS–OH, FEVE, and PEG (Table S1). Subsequently, a series of fluorocarbon composite coatings is fabricated by modifying the content of polyaspartic ester and K-WC (Table S2). Figure 1a displays the Fourier transform infrared spectrum (FTIR) of pristine materials for amphiphilic polymer synthesis. PDMS–OH shows Si–CH₃ peaks at 798 cm^{−1} and 1255 cm^{−1}, while FEVE exhibits a C–F peak at 1210 cm^{−1}. A broad –OH stretching band (3359–3657 cm^{−1}) appears in PEG, PDMS–OH, and FEVE. HDIT displays a characteristic –NCO peak at 2270 cm^{−1}. The FTIR spectra of HP, HPF, and HPFD showed peaks at 3338 cm^{−1} (N–H stretching) and 1535 cm^{−1} (N–H bending), confirming successful prepolymer synthesis (Figure 1b). The absence of peaks near 2270 cm^{−1} in PHN, PFN, PFDN, PFNK, and PFDNK indicated complete –NCO consumption (Figure 1c). 798 cm^{−1} and 1255 cm^{−1} were assigned to Si–CH₃ vibrations in PDMS–OH, while 1210 cm^{−1} and 1045 cm^{−1} corresponded to C–F and C=W vibrations, respectively. The survey spectrum of PHN, PFN, PFDN, and PFDNK (Figure S4a) confirms the presence of C, N, O, F, Si, and W. The C 1s high-resolution spectrum (Figures 1d–e and S4b) shows peaks at 289.2, 288.5, 286.3, 284.8, 283.7, and 282.9 eV, corresponding to C=O, C–F, C–N, C–C, C–Si, and C–W, respectively. In PHN and PFDN, C–N, C–C, and C=O bonds are attributed to urethane formation, while C–F and C–Si in PFDN result from FEVE and PDMS–OH incorporation. The introduction of K-WC combined with the surface migration of Si–O groups leads to the disappearance of C=O and the formation of C–W. Si 2p spectra (Figure S4c) reveal Si–C and Si–O bonds, with PFDN containing 4% silicon (Figure S5a). The above data successfully confirm the chemical structures of amphiphilic polymer coatings. Differential scanning calorimetry (DSC) was used to analyze the glass transition temperature (*T*_g) of PFDN. The results show that incorporating FEVE and PDMS–OH significantly decreases the *T*_g (Figure 1f), attributed to the high bonding energy of Si–O bonds in silicone oil, enhancing molecular chain mobility at low temperatures. Additionally, the addition of K-WC slightly increases the *T*_g of PFDNK due to strengthened physical cross-linking from enhanced hydrogen bonding, improving the coating's barrier protection across varying temperatures.

Surface wettability significantly impacts antifouling and anti-icing properties. The water contact angle (WCA) of a series of amphiphilic liquid-like fluorocarbon coatings is systematically evaluated. The FEVE exhibits a WCA of 87.5 ± 1.3° (Figure 1g). Notably, the incorporation of PEG significantly enhances surface hydrophilicity, resulting in a reduced WCA of 52.3 ±

1.2°. In contrast, the introduction of silicon–oxygen chain segments and K-WC components substantially increases surface hydrophobicity, with WCAs of 85.8 ± 2.6° and 88.1 ± 0.9° for PFDN and PFDNK, respectively. Surface energy (SE) calculations, derived from contact angle measurements using water and diiodomethane (Figure S5b), reveal SE values of 28.3 mJ·m^{−2} and 29.2 mJ·m^{−2} for PFDN and PFDNK (Figure 1g), which are significantly lower than those of commercial FEVE and PHN. The characterization data and analysis of the K-WC component (Figures S6–S7) are provided in the Supporting Information, which shows excellent photothermal conversion of K-WC (AM 1.5, 62.57 °C in Figure S8, photothermal conversion in 84.3%) endows both PFNK and PFDNK with remarkable photothermal properties. The surface temperatures of these coatings reach 44.04 and 47.67 °C, respectively, after 15 min of irradiation (Figure 1h), demonstrating excellent thermal stability.

The extensive application of FEVE in marine environments is significantly constrained by its mechanical deficiencies, particularly the formation of surface microcracks caused by ice impact. These microcracks are further aggravated by seawater erosion, resulting in the deterioration of the coating's intrinsic properties.³² Consequently, enhancing the mechanical robustness is crucial for ensuring the long-term durability of FEVE in marine settings. To address this challenge, a polyurethane/polyurea-modified FEVE composite enhances tensile strength, while incorporating inorganic K-WC as a bridging agent facilitates PFDNK formation—a material exhibiting superior mechanical properties.²³ As shown in Figure 1i, the adhesion strength of PHN reaches 5.43 ± 0.21 MPa after urea bond formation, significantly surpassing that of commercial FEVE. This enhancement can be attributed to the strong hydrogen bonding interactions between the N–H bonds in the coating and the steel substrate. However, the incorporation of fluorocarbon polymers and the increase in silicone content lead to a notable reduction in adhesion strength, with PFN and PFDN exhibiting values of 4.37 ± 0.26 MPa and 3.62 ± 0.20 MPa, respectively. This decline suggests that the conformational chain structure formed by fluorocarbon and silicon–oxygen chain segments hinders the engagement of N–H bonds with the substrate. Remarkably, PFDNK demonstrates improved adhesion (3.90 ± 0.18 MPa), compared to reported adhesion strength values of modified fluoro-silicon polymers. PFDNK demonstrates moderately high adhesion performance within similar modified fluorocarbon material systems (Table S3). This indicates its robust interfacial bonding capability with substrates, suggesting significant potential for practical implementation.³³ Concurrently, the coating hardness increases with the addition of fluorocarbon and silicon–oxygen chain segments (Figure 1i), with PFDNK achieving the highest hardness value of 91.6 ± 1.3 HA. The stress–strain behavior of the coatings is presented in Figure 1j. The inclusion of soft segments, comprising siloxane (PDMS–OH) and polyol (PEG) components, significantly enhances strain capacity, while benzene rings and hydrogen bonds contribute to increased stress capacity. The formation of cross-linked networks between surface sulfhydryl groups on the tungsten carbide and PFDN further improves stress properties, which PFDNK exhibits exceptional mechanical performance, with a maximum elongation at break of 608% and a stress level of 12.94 MPa. Cyclic tensile testing (Figure 1k) is performed on PFDNK at 100% strain, with energy dissipation calculated from the hysteresis curve area (Figure 1l). A significant

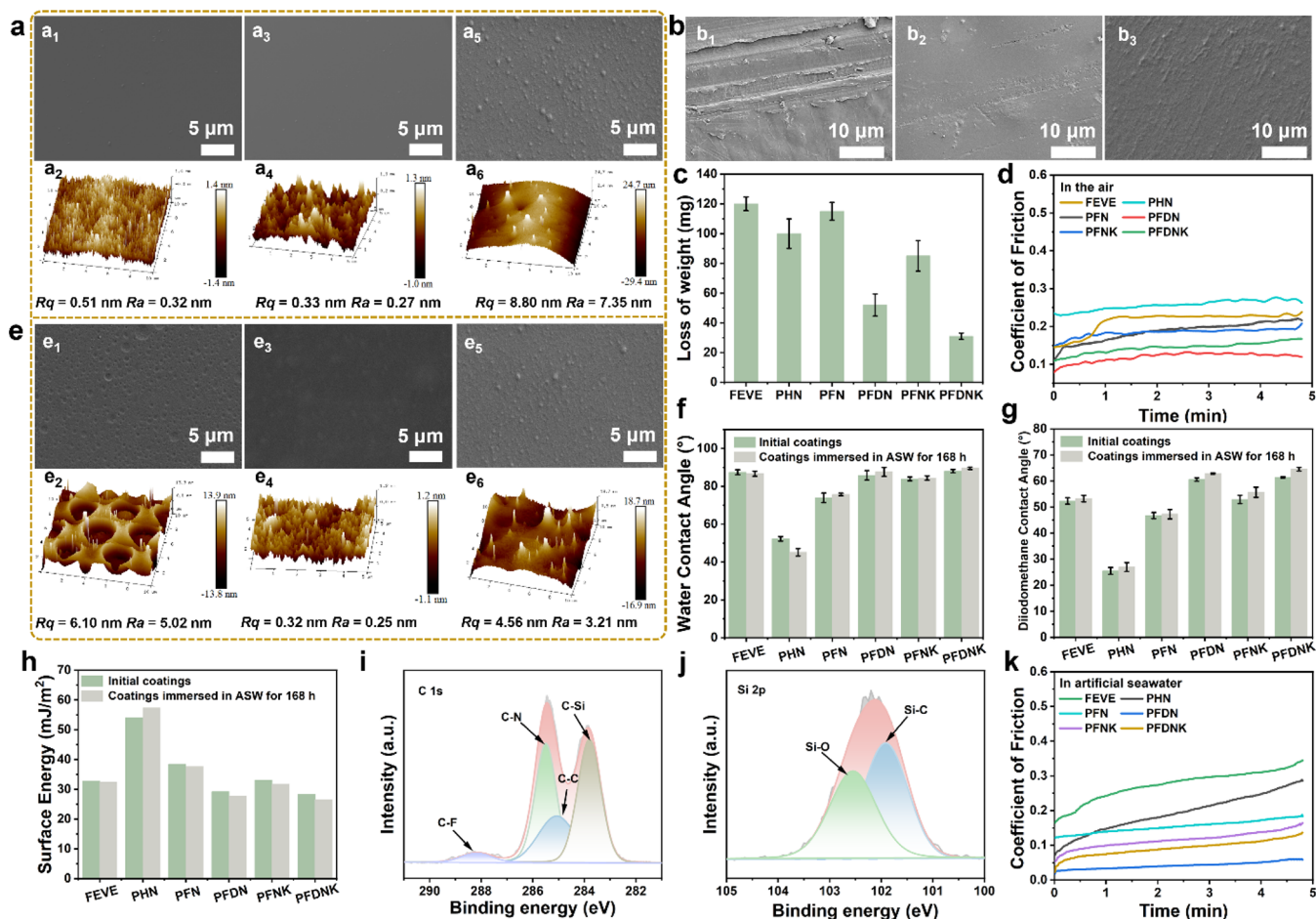


Figure 2. Surface property of the PHN, PFN, PFDN, and PFDNK. (a) SEM and AFM image (a_1 , a_2) PHN, (a_3 , a_4) PFDN, (a_5 , a_6) PFDNK. PHN, PFDN, and PFDNK after 500 Taber abrasion rounds (b) SEM image; (c) weight loss; (d) friction coefficient. (e) SEM and AFM image of coatings immersion in ASW 168 h (e_1 , e_2) PHN, (e_3 , e_4) PFDN, (e_5 , e_6) PFDNK. Different coatings immersed in ASW for 168 h (f) water contact angle, (g) diiodomethane contact angle, (h) surface energy. High-resolution XPS spectra after immersion in ASW for 168 h of PFDN (i) C 1s, and (j) Si 2p. (k) Underwater coefficient of friction.

hysteresis is observed in the first cycle, followed by a substantial reduction in energy dissipation during the second cycle. The energy dissipation in cycles 3–10 shows minimal variation compared to the second cycle, indicating that the majority of hydrogen bonds are disrupted during the initial cycle and cannot fully recover within the testing time frame.³⁴ Comparative analysis with other coatings (Figure S9) reveals that PFDNK exhibits lower energy dissipation and demonstrates rapid shape recovery postdeformation, making it particularly suitable for complex marine environments.

2.2. Surface Properties and Roughness. Surface properties, encompassing both surface morphology and roughness, are critical determinants of antifouling and anti-icing performance. Furthermore, PFDNK demonstrates a notable capacity to reduce surface friction, thereby augmenting its protective efficacy and stability in challenging marine environments. As illustrated by the SEM and AFM images in Figures 2a and S10, all coating surfaces display uniform morphology. The surface roughness of the coatings exhibits a progressive reduction with increasing concentrations of fluorocarbon and silicon–oxygen chain segments. However, a marginal increase in roughness is observed upon the addition of K-WC, with the following specific R_a values recorded: R_a (PHN) = 0.32 nm, R_a (PFDN) = 0.27 nm, and R_a (PFDNK) = 7.35 nm. Following 500 Taber wear cycles under a 500 g load,

SEM analysis reveals distinct wear patterns across the coatings. PHN (Figure 2b₁) displays pronounced wear tracks, while PFDN (Figure 2b₂) and PFDNK (Figure 2b₃) exhibit significantly reduced wear tracks, underscoring exceptional stability and superior wear resistance. Quantitative analysis of weight loss (Figure 2c) further corroborates these findings, with PFDNK exhibiting a minimal mass loss of 31.05 ± 2.01 mg. Additionally, friction coefficient measurements (Figure 2d) reveal that PFDN possesses the lowest coefficient of friction at 0.12. Although PFDNK exhibits a slightly higher coefficient of friction (0.16) due to its increased surface roughness, it still maintains superior low-friction surface properties.

To assess the alterations in surface properties of amphiphilic liquid-like fluorocarbon polymer composite coatings in a marine environment, all coatings are subjected to a 168 h immersion in artificial seawater (ASW). Postimmersion SEM analysis (Figures 2e and S11) reveals significant pitting on PHN, attributable to water absorption through its hydrophilic interface, whereas other coatings maintain their surface integrity. AFM measurements demonstrate increased surface roughness for PHN ($R_a = 5.02$ nm), contrasting with reduced roughness values for PFDN ($R_a = 0.25$ nm) and PFDNK ($R_a = 3.21$ nm). This differential behavior is primarily attributed to the inherent flexibility and surface-enrichment characteristics

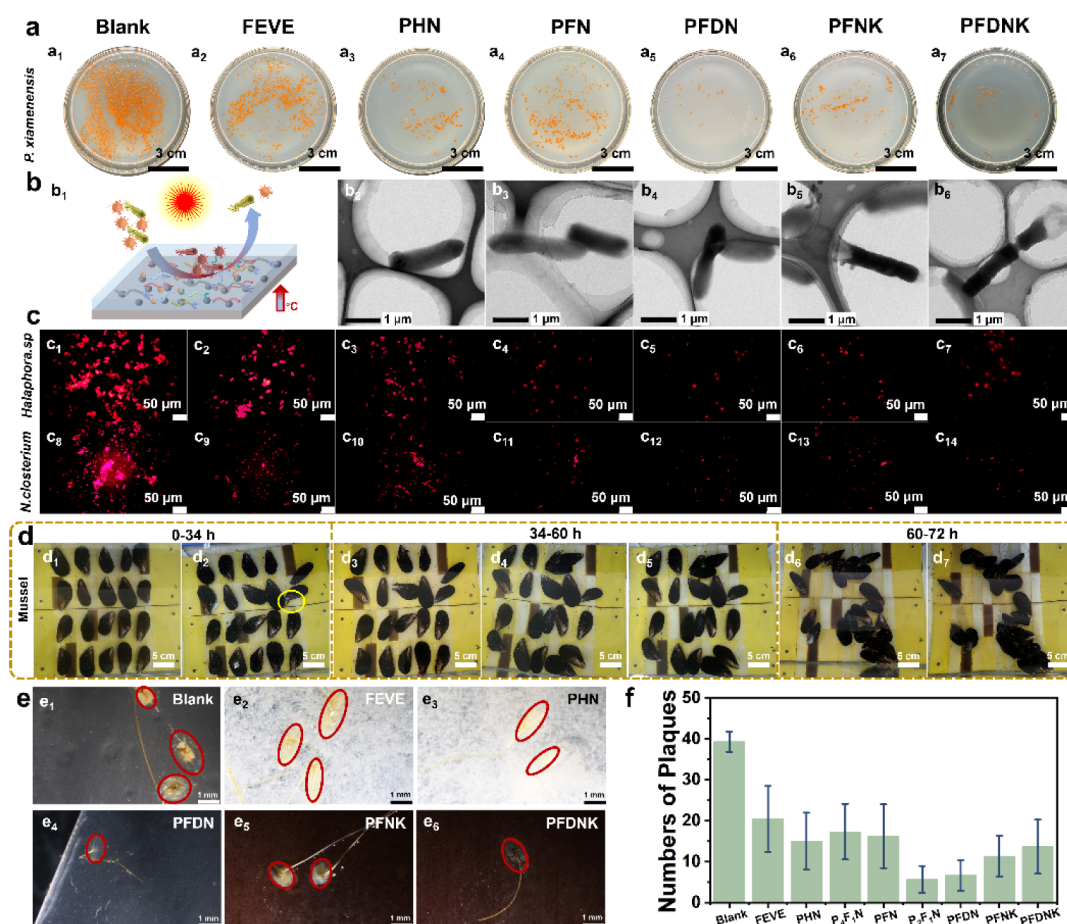


Figure 3. Antifouling property of PHN, PFN, PFDN, and PFNK. (a) *P. xiamenensis* colonial morphology image grown on solid medium. (b) Schematic of the sterilization activity of PFDNK (b₁); TEM images of *E. coli* after exposure to simulated sunlight on different coating surfaces (b₂–b₆). (c) Fluorescence microscope images of algae adhered to different surfaces of *Halamphorasp.* in 7 days and *N. closterium* in 15 days. (d) The locations of mussels at different periods in 72 h. (e) Microscopic image of mussels adhering plaques to different surfaces. (f) The number of mussel-adhesive plaques on different surfaces.

of siloxane chains in PFDN and PFDNK coatings. PFN, PFDN, PFNK, and PFDNK exhibit a gradually increasing water WCA and a decreasing surface energy SE trend, as demonstrated in Figure 2f–h. This phenomenon likely arises because each PEG chain contains two reactive hydroxyl groups (bifunctional) that fully participate in cross-linking, thereby anchoring within the polyurethane/polyurea-modified fluorocarbon matrix. After immersion in ASW, hydrophobic PDMS–OH segments exhibit increased relative exposure at the coating interface, thereby elevating water contact angles. Notably, PFDNK demonstrates optimal performance with an SE of $26.54 \text{ mJ}\cdot\text{m}^{-2}$, attributable to the low SE characteristics of siloxane chains. According to the Baier curve,³⁵ the SE range ($20\text{--}30 \text{ mJ}\cdot\text{m}^{-2}$) confers superior fouling-release properties, while commercial FEVE lost its antifouling capability after 168 h immersion ($\text{SE} = 32.42 \text{ mJ}\cdot\text{m}^{-2}$). XPS analysis reveals a 7.1% surface Si enrichment in PFDN (Figures 2i–j and S12), corroborating siloxane chain migration and explaining its exceptional underwater stability. Figure 2k shows the friction coefficient of the coating in artificial seawater. It can be seen that the friction coefficient of PFDNK in ASW is approximately 0.1, which is 0.06 lower than that in air. PFDN shows the lowest friction coefficient of about 0.07. This phenomenon of reduced friction coefficient underwater is since the amphiphilic liquid-phase fluorocarbon polymer system is

composed of rigid segments (FEVE) characterized by dense hydrogen bonds and flexible segments (PDMS–OH) with lower hydrogen bond content, which are prone to phase separation.³⁶ Under the polar effect of water, the hydrogen bonds in the rigid chain segments are significantly dissociated, and the soft chain segments tend to diffuse to the surface of the coating. The coating formed by grafted PDMS–OH chain segments, which has the characteristic of surface migration, can form a liquid-like layer on the coating surface due to the highly flexible and mobile polymer chains, thereby exhibiting inherent smoothness,^{37,38} and thus presents the phenomenon of low friction coefficient on the coating surface. Collectively, these results demonstrate that PFDNK possesses stable surface characteristics and outstanding wear-resistant properties, making it highly suitable for applications in demanding marine environments.

2.3. Antifouling Performance. Evaluation of marine antifouling performance by investigating the adhesion behavior of proteins, bacteria, algae, and mussels on coating surfaces. The surface-adsorbed protein concentration is evaluated using the Folin-phenol method (also known as the Lowry method),³⁹ with the measured standard curve present in Figure S13a. The amphiphilic liquid-like fluorocarbon coatings demonstrate significantly reduced adsorption of bovine serum albumin (BSA). In particular, PFDN and PFDNK exhibit

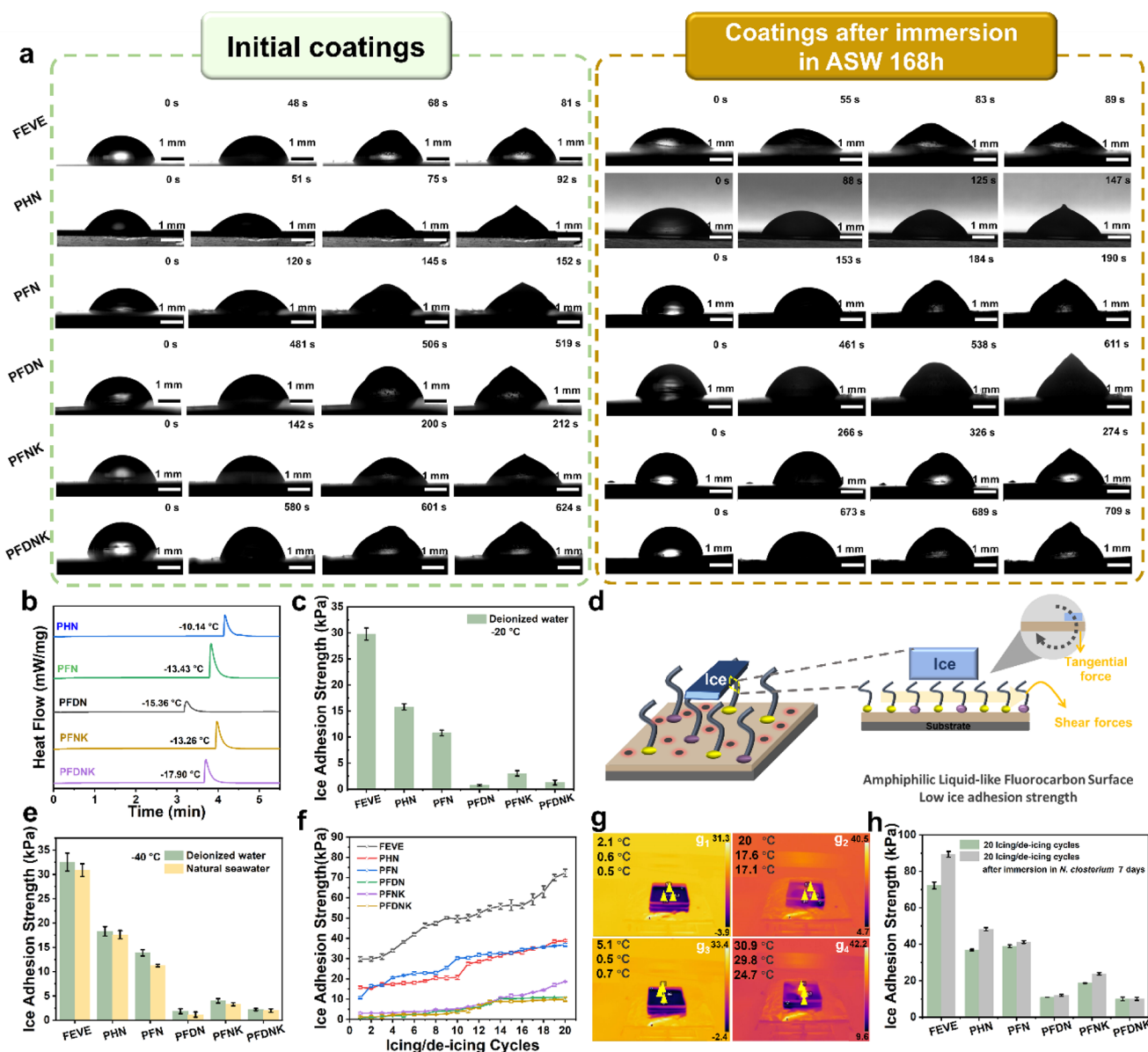


Figure 4. Anti-icing property of PHN, PFN, PFDN, and PFNK. (a) Photographs of individual water droplets frozen process on different surfaces. (b) DSC curves of supercooled water droplets placed into the crucible attached with different surfaces. (c) The adhesion strength between ice and various surfaces by CAT at -20°C . (d) Schematic illustration of the deicing mechanisms on amphiphilic liquid-like fluorocarbon surfaces. (e) The strength of ice adhesion to different surfaces with ice formed by freezing natural seawater and deionized water at -40°C . (f) Ice adhesion strength on different surfaces after 20 icing/deicing cycles. (g) Thermal infrared images (the temperature values in the upper-left region represent measurements taken at three randomly selected locations on each coating surface, as marked by yellow triangles) at 0 and 15 min of ice-covered PFNK (g_1 – g_2) and ice-covered PFDNK (g_3 – g_4). (h) Ice adhesion strength after 20 icing/deicing cycles on different surfaces after immersion in *N. closterium* for 7 days.

remarkably low BSA adsorption of only $0.75\text{ mg}\cdot\text{cm}^{-2}$ and $0.72\text{ mg}\cdot\text{cm}^{-2}$, respectively, which is substantially lower than that of blank glass slides ($8.33\text{ mg}\cdot\text{cm}^{-2}$) and commercial FEVE ($12.31\text{ mg}\cdot\text{cm}^{-2}$) (Figure S13b). These results indicate that the introduction of both fluorocarbon and siloxane segments in PFDNK, combined with its lower surface roughness, effectively suppresses protein adhesion.

Taking glass slides as blank controls and commercial FEVE and PHN as reference samples, the antibacterial properties of amphiphilic liquid-like fluorocarbon coatings are compared under both light and dark conditions. As shown in Figures 3a and S14, all coatings exhibit significantly reduced bacterial colony counts compared to the blank glass slides and

commercial FEVE, with PFDNK demonstrating the most pronounced effect. In dark conditions, PFDNK achieves antibacterial ratios of $95.3 \pm 5.6\%$ against *Escherichia coli* (*E. coli*), $92.1 \pm 3.3\%$ against *Staphylococcus aureus* (*S. aureus*), and $89.3 \pm 5.6\%$ against *Pseudomonas xiamenensis* (*P. xiamenensis*) (Figure S15a). Under light exposure, the antibacterial ratios of PFDNK increase to $98.3 \pm 4.9\%$ for *E. coli*, $97.1 \pm 3.3\%$ for *S. aureus*, and $98.6 \pm 6.1\%$ for *P. xiamenensis* (Figure S15b). The primary reason for this enhancement is the introduction of K-WC, which endows PFDNK with photothermal conversion sterilization capability under illumination. As depicted in Figure 3b₁, the temperature rise on the PFDNK surface effectively reduces and kills surface-adhered bacteria. Fur-

thermore, TEM images of *E. coli* incubated with PHN, PFN, and PFDN under light exposure (AM 1.5) (Figures 3b₂–4b₆) reveal that the bacterial morphology remains intact and viable. In contrast, *E. coli* cocultured with PFNK and PFDNK exhibit disrupted cellular structures, confirming the superior bactericidal effect of these coatings under light conditions.

Figure 3c presents fluorescence microscopy images of blank glass slides, commercial FEVE, and amphiphilic liquid-like fluorocarbon coatings after immersion in *Halophora yongxingensis* (*Halophorasp.*) and *Nitzschia closterium* (*N. closterium*) for 7 and 15 days, respectively. Over time, the number of adhered algae gradually increases on the blank glass slides and commercial FEVE. In contrast, the adhesion of *Halophorasp.* and *N. closterium* to the amphiphilic liquid-like fluorocarbon coatings is significantly lower, particularly for PFDN and PFDNK, which contain silicon–oxygen chain segments. After 7 days of immersion in *Halophorasp.*, the antialgae adhesion ratios are $89.0 \pm 1.5\%$ for PFDN and $85.8 \pm 3.6\%$ for PFDNK. Following 15 days of immersion in *N. closterium*, the antialgae adhesion ratio further improves to $92.4 \pm 3.9\%$ for PFDN and $90.7 \pm 4.3\%$ for PFDNK (Figure S16). Additionally, the absorbance of *N. closterium* cocultured with each coating exhibits an increasing trend over time (Figure S17a), indicating that the prepared amphiphilic liquid-like fluorocarbon coatings do not inhibit algae growth. Simultaneously, zebrafish—a well-established model organism for ecotoxicity assessment—are employed to evaluate nontarget organism safety.^{40,41} As depicted in Figure S17b, 15 zebrafish that are cocultured with PFDN and PFDNK still survived and maintained behavioral activity, which is consistent with the blank control group. This suggests that the coatings are environmentally friendly and function primarily by preventing biofouling adhesion rather than exerting biocidal effects.

The mussel selective adhesion experiment is systematically conducted to evaluate the adhesion behavior of mussels on amphiphilic liquid-like fluorocarbon polymer coatings. Different samples are randomly arranged on mosaic test panels, with the configuration in Figure 3d corresponding to that in Figure S18a (additional arrangements are shown in Figure S18b–c). During the initial 34-h period, the mussels actively extend their feet to explore various coating surfaces. Between 34 and 60 h, they tend to aggregate. In the final 12 h, the mussels demonstrate distinct clustering behavior with only occasional exploratory activities. Notably, both the number and area of mussel byssus plaques on PFDN and PFDNK are significantly smaller than those on blank glass slides, commercial FEVE, and PHN (marked by red circles in Figure 3e), indicating substantially reduced mussel adhesion to PFDN and PFDNK. Consequently, after 72 h, the mussels preferentially aggregate on the glass slides and commercial FEVE. Quantitative analysis from three independent replicates reveals the following average byssus plaque counts: 39.3 ± 2.5 (blank glass slides), 20.5 ± 8.0 (FEVE), 15.0 ± 7.0 (PHN), 16.3 ± 7.8 (PFN), 6.7 ± 3.8 (PFDN), 11.3 ± 5.0 (PFNK), and 13.7 ± 6.6 (PFDNK) (Figure 3f). These results collectively demonstrate that the amphiphilic liquid-like fluorocarbon coatings effectively repel the adhesion of diverse fouling organisms regardless of their species or size.

2.4. Anti-icing Performance. Given the intricate marine environment, high-performance coatings capable of withstanding alternating temporary-cold marine zonal conditions must demonstrate not only exceptional antifouling properties

but also outstanding anti-icing characteristics. To evaluate the anti-icing performance of amphiphilic liquid-like fluorocarbon coatings, we systematically investigate their deicing capabilities under various temperature conditions and different ice formation types. Figure 4a reveals that water droplet freezing times (T_d) on PHN, PFN, and PFDN are 92, 152, and 519 s, respectively, all significantly longer than that on commercial FEVE (81 s). This demonstrates that the incorporation of hydrophilic PEG and highly mobile siloxane segments can effectively delay ice formation. Remarkably, PFDNK exhibits approximately 8-fold longer T_d compared to commercial FEVE, and also outperforms PFDN, which enhancement suggests that K-WC addition improves the anti-icing performance of amphiphilic liquid-like fluorocarbon coatings, potentially through increasing water contact angle. Based on previous analysis, significant migration of siloxane segments is observed in the amphiphilic liquid-like fluorocarbon coatings after ASW immersion. Subsequent T_d evaluation shows prolonged freezing times for all coatings (except FEVE) compared to their pristine states. PFDN displays the greatest T_d increase (92 s), PFDNK maintains the longest T_d (709 s). DSC analysis (Figure 4b) demonstrates crystallization points below $-10\text{ }^\circ\text{C}$ for all coatings, with PFDNK reaching $-17.9\text{ }^\circ\text{C}$, which are attributed to PEG and siloxane segments can strongly bind water molecules, effectively inhibiting hydrogen bond network formation.⁴² Consequently, PFDNK demonstrates the longest retention duration of T_d . The coating's heat flux density (Q) provides further mechanistic insight into ice delay, as expressed in eq 1.⁴³

$$Q = \lambda A \frac{d(T_{\text{base}} - T_{\text{top}})}{d\delta} \quad (1)$$

where λ represents the thermal conductivity coefficient, A denotes the solid–liquid contact area, T_{base} and T_{top} correspond to the temperatures at the coating's bottom and top surfaces, respectively, and δ indicates the coating thickness. $\frac{d(T_{\text{base}} - T_{\text{top}})}{d\delta}$ represents the temperature change rate of the coating, which is primarily influenced by thermal conductivity and contact area. The siloxane segments are densely distributed on the coating surface, functioning as a liquid-like thermal barrier that effectively traps air molecules, consequently reducing the thermal conductivity of PFDNK. Furthermore, compared to FEVE, PFDNK exhibits a larger water WCA, indicating a reduced solid–liquid contact area. The increased thermal resistance and decreased contact area collectively impede heat transfer, thereby enhancing the anti-icing performance and prolonging the freezing time.

Compared to the horizontal push measurement, which exhibits a strong dependence on the contact position of the force probe, resulting in a nonuniform stress distribution at the ice/coating interface, the centrifugal adhesion test (CAT) generates a consistent interfacial stress distribution.⁴⁴ We employ the CAT to assess the ice adhesion strength systematically, whose schematic diagram is presented in Figure S19a. As illustrated in Figure 4c, the CAT measurements reveal ice adhesion strengths of $29.8 \pm 0.9\text{ kPa}$ for commercial FEVE, $15.8 \pm 0.5\text{ kPa}$ for PHN, $10.8 \pm 0.4\text{ kPa}$ for PFN, $0.7 \pm 0.1\text{ kPa}$ for PFDN, $3.0 \pm 0.1\text{ kPa}$ for PFNK, and $1.3 \pm 0.5\text{ kPa}$ for PFDNK. This reduction primarily stems from the interfacial separation effect of the amphiphilic liquid-like fluorocarbon system, which forms a lubricating layer that transforms the solid–liquid interface (substrate–water) into a liquid–liquid

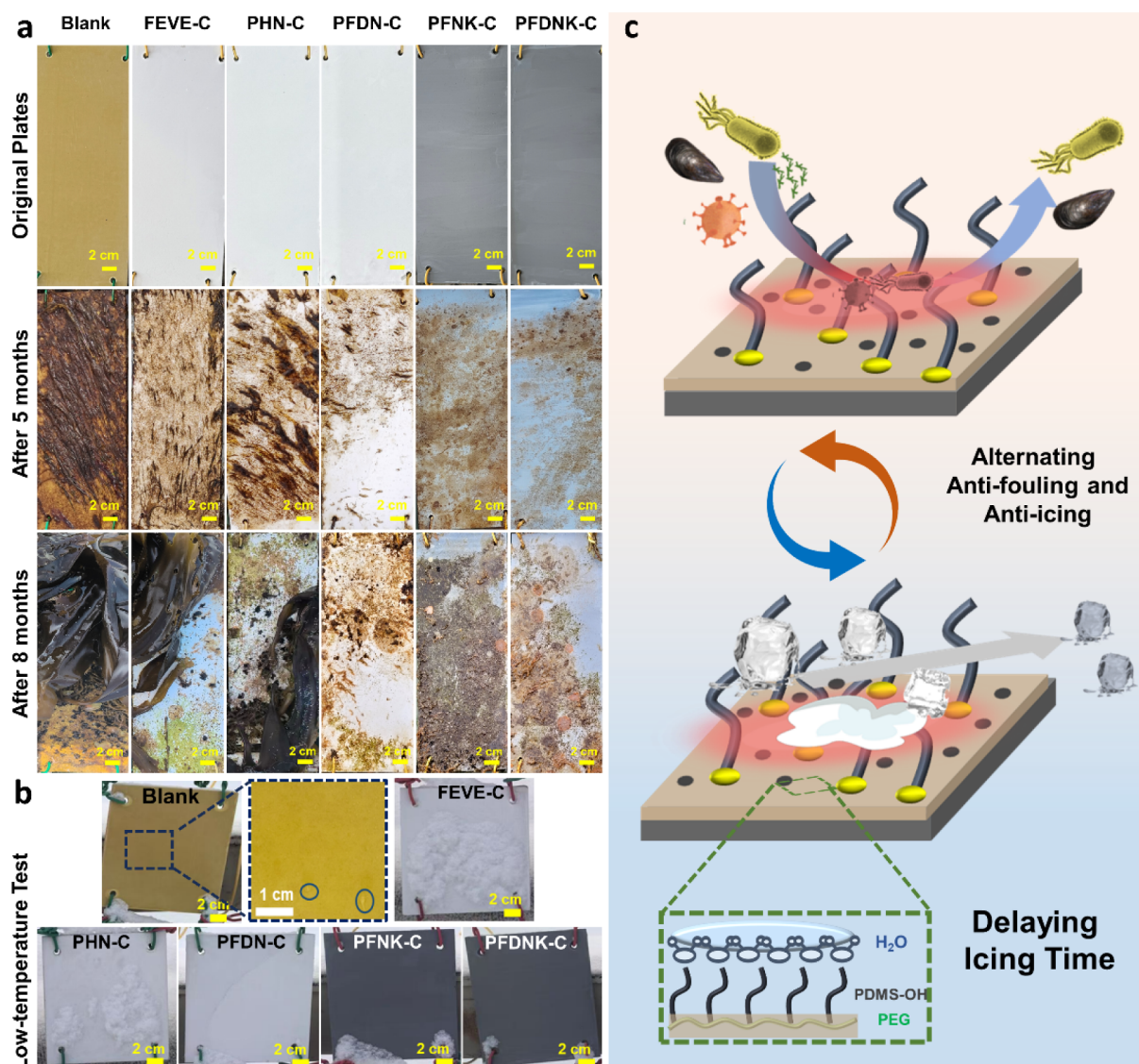


Figure 5. Marine field test of PHN, PFN, PFDN, and PFNK. (a) Different surfaces in the Yellow Sea for 240 days (2024.11–2025.07). (b) Digital photographs of the ice/snow accumulation on various surfaces at an average temperature of -25°C for ≈ 120 days. (c) The antifouling/anti-icing mechanism diagram of PFDNK-C in the alternating temperature and low-temperature marine environment.

interface (liquid-like coating-water). This interface modification reduces surface roughness and prevents direct ice-substrate contact. Consequently, the interfacial transition mitigates the increase in ice adhesion caused by mechanical interlocking.⁴⁵ Meanwhile, the amphiphilic liquid-like fluorocarbon coatings (e.g., PFDN and PFDNK) exhibit more uniform interfacial stress distribution during deicing, facilitating effortless ice removal,^{44,46} as evidenced in Figure 4d. Figure 4e presents a comparative analysis of ice adhesion strengths for deionized water and natural seawater at -40°C . While reduced temperature increased the adhesion strength of deionized water ice, all coatings exhibit lower adhesion with natural seawater ice: PHN (17.6 ± 0.9 kPa), PFN (11.3 ± 0.9 kPa), PFDN (1.2 ± 0.5 kPa), PFNK (3.3 ± 0.3 kPa), and PFDNK (1.9 ± 0.4 kPa). The presence of electrolyte ions (Na^+ , Cl^- , Mg^{2+} , etc.) in seawater disrupts crystal formation, creating localized stress defects that compromise mechanical integrity and interfacial adhesion.^{47,48} Remarkably, the amphiphilic liquid-like fluorocarbon coatings maintain relatively stable adhesion after 20 icing/deicing cycles (Figure 4f).

PFDNK demonstrates exceptional durability with only an 8.7 kPa increase (final value: 10.0 ± 0.9 kPa), while PFDN shows a 10.2 kPa increase (final value: 10.9 ± 0.1 kPa). In stark contrast, commercial FEVE exhibited a substantial 42.4 kPa increase, reaching 72.3 ± 1.5 kPa after 20 cycles, highlighting the superior cyclic stability of amphiphilic liquid-like fluorocarbon coatings. The ice adhesion strength is reduced below the critical threshold (<10 kPa), enabling spontaneous ice removal under natural forces such as gravity, wind, or vibration.⁴⁹ SEM analysis after 20 icing/deicing cycles reveals distinct frictional marks on FEVE and PHN, while only minor scratches are observed on PFN, PFDN, PFNK, and PFDNK (Figure S19b–g). However, the scratch formation leads to increased surface roughness of the coating. Microscopically rough surfaces not only provide larger cavities accommodating potential ice anchoring points and extensive mechanical interlocking, but also expand the ice contact area, thereby increasing the true interfacial contact area,⁴⁴ which leads to an increase in ice adhesion strength after 20 icing/deicing cycles. The photothermally active K-WC component conferred ice-

melting functionality to PFNK and PFDNK, as quantitatively demonstrated in Figures 4g and S20. PFDNK surfaces reach 8.21 °C within 120 s of AM 1.5 irradiation despite 2 mm ice coverage, ultimately achieving thermal equilibrium at 31.33 °C after 15 min of continuous illumination. Consequently, the synergistic interaction of photothermal ice melting and low ice adhesion endows PFDNK with exceptional ice resistance.

The coatings' stability under alternating antifouling/anti-icing conditions is assessed through 7-day *N. closterium* immersion followed by 20 icing/deicing cycles (Figure 4h). While commercial FEVE shows substantial ice adhesion increase (~ 17.1 kPa to 89.3 ± 1.6 kPa), the amphiphilic liquid-like fluorocarbon coatings maintain remarkable stability, particularly PFDN (~ 1.0 kPa to 11.9 ± 0.5 kPa) and PFDNK (~ 0.1 kPa to 10.0 ± 0.8 kPa), confirming their dual functionality.

2.5. Marine Field Test. To validate the practical performance of the amphiphilic liquid-like fluorocarbon composite coating in a real marine environment, a 240-day (from November 2024 to July 2025) marine field test was conducted near Dachangshan Island in the Yellow Sea (Figure 5a). During this period, substantial fouling by seaweed and kelp is observed on the bare epoxy panels. The commercial FEVE-C surface also exhibits significant biofouling coverage, accompanied by coating detachment. Similarly, substantial attachment of seaweed and kelp is observed on the PHN-C coating (prepared from non-cross-linked FEVE and PDMS-OH). In contrast, the PFDN-C and PFDNK-C coatings demonstrate effective resistance to macrofouling adhesion. After 8 months of immersion, only minor seaweed attachment is present on the PFDN-C and PFDNK-C surfaces. Notably, the PFDNK-C coating demonstrates superior real-world marine performance under low-temperature and fluctuating temperature conditions, exhibiting no damage, no discernible surface delamination, and minimal bioaccumulation. Additionally, all coatings were placed in an outdoor environment with an average temperature of -25 °C (approximately 120 days), and the formation of ice or frost was photographically documented (as shown in Figures S21 and 5b). The blank epoxy substrate exhibits significant ice accumulation (Video S1), while FEVE-C and PHN-C demonstrate substantial frost and snow deposition. In contrast, the PFDNK-C surface shows no frost or snow accumulation, indicating its exceptional anti-icing performance in real-world low-temperature environments.

Liquid-like coatings not only impart low surface adhesion but also exhibit Couette-like flow behavior under shear stress.⁵⁰ This characteristic enables them to mimic liquid lubricants during shear, forming a liquid-like interface that facilitates liquid slip.⁵¹ Consequently, liquid-like coatings achieve significant drag reduction by minimizing velocity gradients and turbulent kinetic energy. Drag reduction testing of the amphiphilic liquid-like fluorocarbon composite coating (Figure S22) reveals that PFDN-C and PFDNK-C exhibit outstanding drag reduction properties, with drag reduction rates of 32.6% and 30.2%, respectively. Furthermore, surface stability under harsh environments is evaluated through high-salinity immersion tests. Samples of PHN-C, FEVE-C, PFDN-C, and PFDNK-C are immersed in ASW (4.1 wt % salinity) for 14 days. As shown in Figure S23, postimmersion analysis reveals distinct surface degradation patterns in microscope images: PHN-C exhibits significant blistering, while FEVE-C shows minor particle leaching of solid fillers. In marked

contrast, the microscope images of PFDN-C and PFDNK-C maintain pristine surface integrity with no visible alterations, demonstrating exceptional stability in high-salinity environments. This robust surface preservation ensures reliable antifouling and anti-icing performance of the coatings.

2.6. Analysis of Antifouling/Anti-icing Mechanisms.

Taking PFDNK-C as an example, the alternating antifouling/anti-icing mechanism of the amphiphilic liquid-like fluorocarbon composite coating in both temperate and low-temperature marine environments is illustrated based on the marine field test results, as shown in Figure 5c. When exposed to fouling organisms in temperate waters, the underwater-enhanced properties of the amphiphilic liquid-like coating facilitate the migration of flexible chain segments (PDMS-OH) to the surface, endowing PFDNK-C with a "non-adhesive" characteristic that effectively inhibits the attachment of various fouling organisms. Simultaneously, the photothermal properties of K-WC enable PFDNK-C to elevate its surface temperature under solar illumination, achieving bactericidal effects. In low-temperature marine environments, PDMS-OH segments densely distributed on the PFDNK-C coating surface form a liquid-like thermal barrier that captures air molecules, effectively reducing thermal conductivity. The resultant increase in thermal resistance impedes heat transfer, thereby prolonging freezing time. The friction-reducing properties are imparted by PDMS-OH further reduce ice adhesion strength, allowing accumulated ice to be easily removed. Additionally, the photothermal conversion capability of PFDNK-C enables solar energy-induced melting of surface ice upon solar irradiation. The fluorocarbon composite coating, synergistically formed by the underwater-enhanced amphiphilic liquid-like layer and photothermal effect, constructs a surface with alternating and durable antifouling and anti-icing properties. Furthermore, the application of protective coatings typically involves large-scale scenarios, making the scalability of synthesis a critical factor. The synthesis of the polyurethane/polyurea-modified fluorocarbon system utilizes a well-established nucleophilic addition reaction, which represents a mature industrial process extensively adopted in polyurethane manufacturing.^{52,53} And the raw materials used are all commodity chemicals readily available commercially. The above demonstrates the strong scalability potential of the polyurethane/polyurea-modified fluorocarbon polymer system presented in this work.

3. CONCLUSION

In conclusion, inspired by the "non-adhesive" strategy of Pacific salmon mucus, we have designed and fabricated an amphiphilic liquid-like fluorocarbon polymer (PFDN) for marine antifouling/anti-icing applications. By integrating K-WC into PFDN, the resulting PFDNK coating demonstrates durable anti-icing performance and stable antifouling capability. The incorporated K-WC endows the coating with exceptional ice-melting capacity and enhances its antifouling performance under illumination, thereby meeting diverse protective requirements. Furthermore, the coating exhibits outstanding mechanical properties, remarkable underwater friction reduction, and strong adhesion strength. And the coating maintains robust surface stability under both high-salinity immersion and turbulent drag reduction testing conditions. In the practical marine field (approximately 240 days) and the real-world low-temperature tests (approximately 120 days), the coating demonstrates significant performance in

antifouling and anti-icing applications. Consequently, this coating system represents a long-term protection strategy for various underwater equipment operating in alternating marine environments with fluctuating temperatures.

4. EXPERIMENTAL SECTION

4.1. Materials. Chemicals were purchased from Tianjin Guangfu, Aladdin, Shanghai Macklin Biochemical, Sigma-Aldrich, etc. and used without any further purification. All solvents were reagent grade and were dried before use according to standard procedures. More details are available in the [Supporting Information](#).

4.2. Characterization and Property Testing. FT-IR spectroscopy (Thermo Nicolet, 500–4000 cm^{-1}) was used to analyze the polymer samples. The mechanical properties of the polymer films were measured by a WDW universal testing machine (Kexin). The thermal properties and water crystallization points were measured by differential scanning calorimetry (TA Instruments, DSC, Q20 system, heating/cooling rate = 5 $^{\circ}\text{C min}^{-1}$, N_2 atmosphere). SEM (Thermo Fisher) and AFM (Bruker Dimension Icon) were performed to obtain the microstructure and morphology information. A contact angle meter (Edmund Optics) was used to detect the contact angle and the change in water to icing over time. More details are available in the [Supporting Information](#).

The sections describing the anti-icing experiments, antibacterial evaluation, antialgae performance evaluation, environmental-friendliness assessment, and mussel choice and settlement assays etc. are presented in the [Supporting Information](#).

4.3. Compound Synthesis. Before use, PEG, PDMS-OH, and FEVE were heated in a dried glass flask at 100 $^{\circ}\text{C}$ under vacuum for 1 h to remove moisture and then cooled to room temperature. A drop of dibutyltin dilaurate ($\approx 20 \mu\text{L}$) was added to a mixture of PEG (20 g, 0.01 mol), FEVE (20 g, 0.015 mol) PDMS-OH (40 g, 0.03 mol), and hexamethylene diisocyanate trimer (80 g, 0.13 mol) in dry xylenes (20.00 mL), and the mixture was stirred at 80 $^{\circ}\text{C}$ for 3 h to obtain the HPFD prepolymer, the specific contents are shown in [Table S1](#). A solution of polyaspartic ester—NH1420 (5.71 g, 0.01 mol) in dry xylenes (6.00 mL) was then added, and the mixture was stirred at room temperature for 5 min to deactivate excess isocyanate groups. After the reaction, the mixture was poured into a tetrafluoroethylene mold or coated on glass, copper, and steel plates, followed by solvent evaporation to obtain PFDN film (thickness: 0.1–0.2 mm) for further measurements. The details of K-WC's synthesis are presented in the [Supporting Information](#). The PHN and PFN followed similar synthetic strategies. PFNK and PFDNK were introduced as K-WC (0.5 g, 3.2 wt %) in PFN and PFDN, respectively ([Table S2](#)). The preparation method for the antifouling paints is detailed in the [Supporting Information](#).

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.5c12461>.

Synthetic formulations of HP, HPF, and HPFD prepolymer; synthetic formulations of PHN, PFN, PFDN, PFNK, and PFDNK; adhesion strength of the PFDNK and previously reported materials; the schematic diagram for the synthesis of HPFD and PFDN;

the synthesis of K-WC; schematic diagram of the PFDNK; high-resolution XPS of PHN, PFN, and PFDN; content of elements in PFDN; XRD, FTIR, UV–vis–NIR spectra of WC and K-WC; TEM and SEM images of WC and K-WC; temperature change of K-WC; cyclic tensile and the corresponding energy dissipation of PHN, PFN, PFNK, and PFDN; SEM images of FEVE, PFN, and PFNK; SEM images of FEVE, PFN, and PFNK coatings immersion in ASW 168 h; content of elements of PFDN immersion in ASW 168 h; adsorption of BSA on different surfaces; morphology images of bacteria on different coatings; antibacterial ratio; antialgae ratio; growth curves of *N. closterium* and environmental toxicity assessment tests of zebrafish; sample distribution diagrams of mosaic tests; device diagram for deicing by the CAT method and SEM images of coatings after 20 cycles of icing/deicing; temperature change of PFNK and PFDNK; digital images of ice and snow buildup on different surfaces; drag reduction rates; microscope images of FEVE-C, PHN-C, PFDN-C, and PFDNK-C immersion in high-salinity simulated seawater ([PDF](#))

Video showing the icing condition on the surface of the blank epoxy substrate ([MP4](#))

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Notes

The authors declare no competing financial interest.

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